Towards Interpretable Protein Structure Prediction with Sparse Autoencoders

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Abstract

Protein language models have revolutionized structure prediction, but their nonlinear nature obscures how sequence representations inform structure prediction. While sparse autoencoders (SAEs) offer a path to interpretability here by learning linear representations in high-dimensional space, their application has been limited to smaller protein language models unable to perform structure prediction. In this work, we make two key advances: (1) we scale SAEs to ESM2-3B, the base model for ESMFold, enabling mechanistic interpretability of protein structure prediction for the first time, and (2) we adapt Matryoshka SAEs for protein language models, which learn hierarchically organized features by forcing nested groups of latents to reconstruct inputs independently. We demonstrate that our Matryoshka SAEs achieve comparable or better performance than standard architectures. Through comprehensive evaluations, we show that SAEs trained on ESM2-3B significantly outperform those trained on smaller models for both biological concept discovery and contact map prediction. Finally, we present an initial case study demonstrating how our approach enables targeted steering of ESMFold predictions, increasing structure solvent accessibility while fixing the input sequence. To facilitate further investigation by the broader community, we open-source our code, dataset, pretrained models, and visualizer.

1 Introduction

Machine learning-based protein structure prediction models have achieved remarkable success by leveraging vast amounts of sequence data (Jumper et al., 2021; Lin et al., 2022), building on the success of previous sequence homology-based methods (Marks et al., 2011; Kamisetty et al., 2013). However, their nonlinear nature obscures how sequence information informs structure prediction.

Previous interpretability studies on transformer-based protein language models (PLMs) have demonstrated PLMs learn structural information despite only training on sequences. Rao et al. (2021) and Vig et al. (2021) showed that the attention map learns to predict residue-residue contacts. Zhang et al. (2024) show PLMs predict structure primarily by memorizing and retrieving patterns of coevolving residues. However, these works have yet to establish causal relationships between internal mechanisms and predictions.

Sparse autoencoders (SAEs) offer a promising direction for understanding language models by learning interpretable linear representations (Templeton et al., 2024; Gao et al., 2024). Prior works (Simon and Zou, 2024; Adams et al., 2025) train SAEs to uncover thousands of biologically interpretable features in ESM2 models, but these studies focused on smaller variants rather than ESM2-3B, the language model underlying ESMFold's structure prediction.

Our work advances protein model interpretability by scaling SAEs to ESM2-3B (the base model for ESMFold) and also adapting Matryoshka SAEs (Nabeshima, 2024; Bussmann

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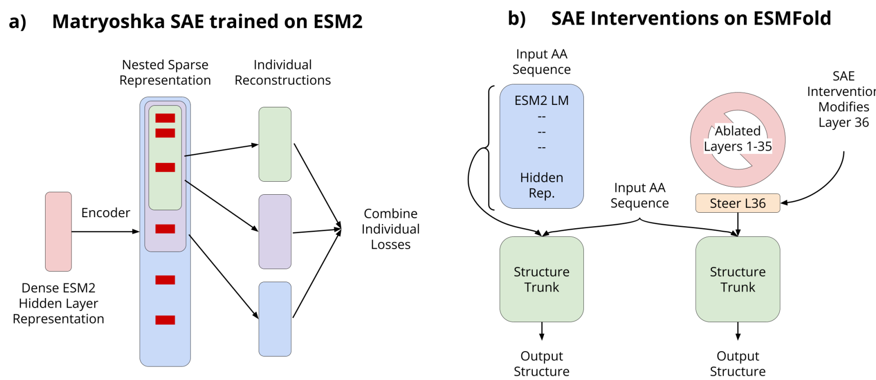


Figure 1: a) Matryoshka Sparse Autoencoder (SAE) architecture for training on ESM2 hidden layer representations, showing nested sparse feature organization. b) SAE intervention framework for ESMFold, comparing normal operation (left) where all ESM2 hidden representations flow to the structure trunk, versus intervention (right) where only a modified layer 36 representation is used while ablating all other layers.

et al., 2024) to learn hierarchically organized features that align with protein structure's multi-scale nature.

We structure the paper as follows: In Section 2, we present scaling SAEs to ESM2-3B and Matryoshka SAEs for hierarchical features. Section 3 shows Matryoshka SAEs match or exceed L1 and TopK in language modeling and structure prediction. Section 4 evaluates ESM2-3B's performance on biological concept discovery and contact prediction. Section 5 presents feature steering to control protein structure properties. Code, models, and a visualizer will be released upon publication.

2 Methods

2.1 Setup

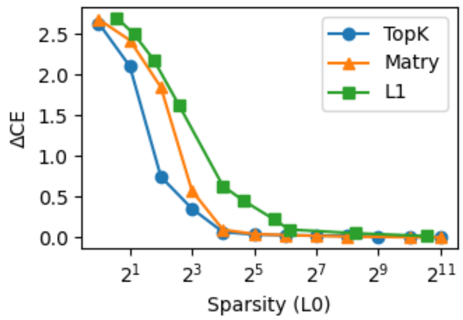
Problem Statement. Given token embeddings $x \in \mathbb{R}^d$ from a transformer-based PLM, our SAE encodes x into a sparse, higher-dimensional latent representation $z \in \mathbb{R}^n$ where $d << n$, and decodes it to reconstruct x by minimizing the L2 loss $\mathcal{L} = |x - \hat{x}|_2^2$. We enforce sparsity on z through methods detailed in Appendix I.1.

Models and Training. We train SAEs on layers 18 and 36 of ESM2 (Lin et al., 2022), focusing on the 3 billion parameter ESM2-3B model which provides representations for ESMFold. During training, embeddings are normalized to enable hyperparameter transfer between layers, with biases scaled during inference following Marks et al. (2024). Our training data consists of 10M sequences randomly selected from UniRef50, constituting 2.5 billion tokens. See Appendix I.3 for details.

2.2 Matryoshka SAEs

Proteins exhibit inherent hierarchical organization across scales, from local amino acid patterns to molecular assemblies. We employ Matryoshka Sparse Autoencoders (SAEs), which learn nested hierarchical representations through embedded features of increasing dimensionality (Fig. 1a) (Nabeshima, 2024; Bussmann et al., 2024).

Our implementation follows Bussmann et al. (2024), Marks et al. (2024), and Gao et al. (2024) in dividing the latent dictionary into nested groups where earlier groups capture abstract features while later groups encode granular details. For mathematical details of the encoding, decoding, and loss computation, see Appendix D.



(a) CE loss reconstruction across sparsity levels for TopK, Matryoshka (Matry), and L1 regularized autoencoders. Matryoshka requires similar or less active latents to achieve good reconstruction.

Comparison	RMSD (Å)
Exp vs. ESMFold (baseline)	3.1 ± 2.5
Exp vs. ESMFold (full ablation)	15.1 ± 5.9
Exp vs. ESMFold (only layer 36)	2.9 ± 2.1
Exp vs. SAE (layer 36)	3.2 ± 2.6
ESMFold vs. SAE (both L36)	3.1 ± 4.4

(b) Backbone RMSD (Å) comparing experimental structures (Exp), ESMFold predictions, and SAE reconstructions. Keeping layer 36 or using SAE preserves accuracy, while full ablation degrades performance.

3 Evaluations on Downstream Loss

We evaluate our SAEs on language modeling and structure prediction for ESM2-3B and ESMFold respectively to assess how well they preserve performance on downstream tasks.

3.1 Language Modeling

Setup. We report the average difference in cross-entropy loss ΔCE between the logits from the original and SAE reconstructed PLM. We evaluate on a heldout test set of 10k sequences randomly sampled from Uniref50 on layer 18 of ESM2.

Results. Figure 2a shows the impact of different autoencoder architectures on downstream language modeling performance across sparsity levels. For architecture details, see Appendix I.2. At low sparsity ($L_0 < 10$), all approaches - TopK, Matryoshka, and L1 regularization - show similar degradation (ΔCE $\approx$ 2.5–3.0). As sparsity increases, TopK and Matryoshka SAEs maintain better performance compared to L1 regularization, with ΔCE approaching 0 for sparsity levels above 100.

3.2 Structure Prediction

Setup. By scaling SAEs to ESM2-3B, we can now evaluate how well our SAE representations preserve ESMFold's structure prediction capabilities. We focus on reconstructing ESM2's hidden representations that feed into ESMFold. Since ESMFold uses representations from all layers but our SAE reconstructs only one layer, we ablate all other layers to isolate reconstruction effects. This ablation maintains performance on the CASP14 test set (see Fig. 2b).

Data. We evaluate structure prediction on the CASP14 dataset, a diverse challenging benchmark held out from ESMFold during training. After filtering for computational constraints and prediction quality (see Appendix F), we analyze 17 protein targets.

Results. Our experiments demonstrate effective preservation of structural information. In Fig. 2b, we see that comparing experimental structures to ESMFold predictions shows an RMSD of 3.1 ± 2.5 Å without ablation. While full ablation of ESM2 embeddings significantly degrades performance (RMSD 15.1 ± 5.9 Å), both keeping only layer 36 (RMSD 2.9 ± 2.1 Å) and using our SAE reconstruction (RMSD 3.2 ± 2.6 Å) maintain comparable performance. The similarity between ESMFold and SAE predictions (RMSD 3.1 ± 4.4 Å) confirms preservation of structural information. We also see that in Fig. 7, with only 8 to 32 active latents, SAEs can reasonably recover structure prediction performance.

These results show our SAE effectively compresses model representations while maintaining both sequence-level and structural prediction capabilities across sparsity levels.